



Review

Update on food safety of monosodium L-glutamate (MSG)



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ABSTRACT

Objective: This evidence-based safety review of the flavor enhancer monosodium L-glutamate (MSG) was triggered by its global use and recent studies expressing some safety concerns.

Methodology: This article obtained information through search of evidence-based scientific databases, especially the US National Library of Medicine NIH.

Results: (A) MSG is a water-soluble salt of glutamate, a non-essential amino acid, normally synthesized in the body and prevalent in protein foods. (B) MSG is utilized world-wide for its “umami” taste and flavor enhancement qualities, (C) the human body does not discriminate between glutamate present in food and that added as seasoning, (D) glutamate metabolism is compartmentalized in the human body without reported ethnic differences, (E) glutamate does not passively cross biological membranes, (F) food glutamate is completely metabolized by gut cells as energy source and serves as key substrate for other important metabolites in the liver, (G) normal food use of MSG is dose-dependent and self-limiting without elevation in plasma glutamate, (H) the recent EFSA acceptable daily intake (30 mg/kg body weight/day) is not attainable when MSG is consumed at normal dietary level, (I) scientists have not been able to consistently elicit reactions in double-blind studies with ‘sensitive’ individuals using MSG or placebo in food.

Conclusion: Based on the above observations (A–I), high quality MSG is safe for all life-cycle stages without respect to ethnic origin or culinary background. MSG researchers are advised to employ appropriate scientific methodologies, consider glutamate metabolism and its normal food use before extrapolating pharmacological rodent studies to humans.

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1. Introduction

Monosodium L-glutamate (MSG) is a sodium salt of glutamic acid – a non-essential amino acid present in all protein foods –

used globally as a food flavor enhancer [1]. MSG safety has been reviewed extensively in the 1970–1990s by several leading organizations and academic expert bodies [2–40]. JECFA [28] declared that a numeric “acceptable daily intake” (ADI) was not necessary for glutamic acids and its salts; a view which SCF [29], FASEB [33] and FDA [41] upheld after further evaluations. Recently, European Food Safety Authority (EFSA, [42]) concluded safety re-evaluation

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and established an acceptable daily intake level of 30 mg/kg body weight.

Moreover, due to its wide use and safety history, the UN Children's Fund and the Federal Government of Nigeria [43,44] identified MSG as one of the potential food vehicles for fortification with micronutrients to curb the menace of micronutrient deficiencies (hidden hunger). Hidden hunger is still plaguing most part of the global communities especially economically developing regions. For example, it has been estimated that more than two billion people are anemic because of iron deficiency alone [45]. Interestingly, MSG could not only become an appropriate fortification vehicle for iron, but its use alone has been associated with an increased hemoglobin levels in humans [46,47].

In spite of this rich source of literature on safety of food-used MSG, several new studies have recently appeared, questioning the substance safe use in the global community [48–50]. Therefore, it is appropriate to review MSG safety facts in the light of the recent pharmacological studies conducted in some parts of the world, so that the future use of the substance in micronutrient fortification is not dissuaded by misinterpreted studies.

1.1. MSG: the flavor enhancer

MSG's origin as a flavor enhancer can be traced back to the Roman Empire's use of glutamate-rich fermented fish condiments, called "Garum" [51], and the East Asian use of seaweed in making local food stocks [14]. The seeds of legumes (African locust bean, castor seeds, soy bean) are fermented as condiments in Africa [52–54]; in Nigeria, traditionally fermented vegetable protein based condiments (dawadawa, iru, ogiri) have been consumed by different ethnic groups for centuries [55]. The increased nutritive value and flavor enhancing properties of these plant proteins was attributed to fermentation and increase in amino acids profile [56,57–59]. All above-mentioned fermented condiments utilize the flavor richness associated with the high level of glutamate.

Chemically, the amino acid glutamate was first isolated by Ritthausen in 1866 from the proteins in wheat. In 1908, a Japanese professor at the Imperial Tokyo University, Kikunae Ikeda, observed that specific sea-weeds contained generous amounts of glutamate [14], which he linked to the basic taste "umami". Soon thereafter, sodium salt of glutamate (MSG) was found to be the most stable and practical form of the commercial use of glutamate. Since 1950s, MSG has been produced via fermentation processes from sugar beet molasses and carbohydrate sources (e.g., corn, sugar beet), similarly to the production of beer, wine or vinegar. The key difference between MSG and other fermented foods is the crystallization and high purity levels of commercial MSG [27,60].

In some parts of West Africa, MSG is locally known as "Maggi white". Indeed, MSG is a white crystalized powder, which is readily soluble in water. It is also soluble in ether but insoluble in alcohol, acetone, benzene, methanol and acetic acid [60]. It has a pH of 6.8–7.2. Importantly for its commercial use in Africa, MSG is affordable, resistant to humidity and very stable in various storage conditions.

Glutamate is the most abundant amino acid in all protein foods (relatively low in animal proteins and high in vegetable proteins) [61]. In numeric terms, glutamate constitutes up to 8–10% of amino acids content in the average dietary proteins [62,63] either in a free form (not part of a protein) or bound form (as one of the amino acids making up protein) [64]. Glutamate occurs in many savory foods like mushrooms, cured ham, cheese, tomatoes, scallops, tuna, green peas, fish and soy sauces, beef, yeast extract, human and cow's milk etc. [64–66]. Among various forms of glutamate, the sodium form (MSG) has the most prominent flavor enhancing capacity and umami potency [67–69]. However, it is notable that there are

other global commercial forms of free glutamate such as soy sauce, hydrolyzed vegetable proteins and autolyzed yeast extract [70,71].

Going beyond the flavor enhancer use of free (non-protein) glutamate, the protein-bound glutamate from meats and plants is either extracted by heating during cooking process, or released from the protein chain in the small intestine due to the actions of digestive enzymes. In the second case, glutamate does not enhance flavor by binding in the taste receptors in the tongue, because its taste properties are disabled by the peptidic bond with other amino acids. As a flavor enhancer, free glutamate and especially the MSG form of free glutamate, is widely used. It is valued by food manufacturers for the ease of use as well as its safety, and by chefs for its ability to enhance and blend natural flavor characteristics [14]. Similar to salt, only small quantity of MSG is required to optimize umami taste and the excess does not make food tastier [72,73]. This indicates that MSG use in food is self-limiting.

Interestingly, MSG does not have a distinct taste of its own but it interacts with food matrices, heightens and intensifies the natural flavor of foods, drawing out hidden flavors and enhancing weak ones [26]. This fact might be the key reason that in all culinary cultures in which MSG has been used, it has been consumed as a constituent of meals, not in isolation.

Food intake data in USA and Europe indicated an average daily consumption of food-added MSG at approximately 0.6 g, with a range of 0.3–1.0 g/day [72,73]. Similarly, Giacometti, [74] and Raiten et al. [63] identified a daily intake of food-added MSG between 0.4 and 0.5 g/day. UK data estimated average daily consumption of MSG at approximately 0.6 g/day with 2.0 g/day in high consumers [75]. In East and Southeast Asian diets, MSG intakes are 2–3 times higher when compared to those reported in Europe; at 1.5–3.0 g/day in Taiwan, 1.1–1.6 g/day in Japan and 1.6–2.3 g/day in South Korea [74,76,77]. In Nigeria, consumption studies documented an average intake of 0.56–1.00 g/day [40,43], which is comparable to the intakes found in both the USA and Western Europe. The wide use of glutamate by both rural and urban households in Nigeria has been attributed to the unique taste it adds to the diet, its convenience, availability and affordability [40]. This wide accessibility has prompted some authors to advocate MSG as a potential vehicle for micronutrient fortification in Nigeria [36,38].

All the above-described ranges of MSG dietary consumption represent but a small fraction (5–10%) of the total daily glutamate intake from various dietary sources [78]. Adult humans ingest between 10.0 and 20.0 g glutamate per day irrespectively of their ethnicity, culinary culture or dietary habits [74,76,79].

1.2. Glutamate in the human gastrointestinal tract

Total free glutamate contained in the adult human body is about 10 g and the highest glutamate content is in the human brain (2.3 g) and muscle (6.0 g) [74]. The human body does not discriminate between naturally occurring glutamate in foods and glutamate added in the form of a seasoning [33]. Hence, from an evolutionary as well as nutritional viewpoint, high quality MSG (MSG that is not adulterated by heavy metals or other substances) does not bring anything new to a human diet. The long evolutionary use of glutamate as a taste signaling molecule implied that the metabolism of glutamate is compartmentalized in the organs and that the glutamate does not passively cross biologic membranes. Less than 5% of the ingested glutamate passes from the gut into the circulation and none transits the placenta or the blood brain barrier [78].

The gastro-intestinal tract is the main barrier to the penetration of ingested MSG into the rest of the mammalian body and mammals have the ability to metabolize very large doses [80]. When glutamate is absorbed from the intestinal lumen, almost all is immediately metabolized by intestinal cells to generate ATP –providing energy necessary for intestinal motility [81–85].

Further metabolism of the remaining small amounts of glutamate occurs in the liver. This process is made of simultaneous transamination of glutamate to alanine and aspartate, as well as a formation of other minor metabolites (α -ketoglutarate, glutamine, γ -aminobutyrate, urea and glutathione), which are later incorporated into intestinal proteins [86]. The remaining amount of the ingested glutamate (<5%) enters into the portal circulation [83,84,87,88]. Therefore, systemic blood levels rise only when pharmacological doses (>12 g) are ingested on an empty stomach to overwhelm the metabolism; or if MSG is administered by parenteral routes [70].

Taken together, the rapid metabolism and use of glutamate in the intestine explains why normal dietary consumption of MSG has no effect on plasma glutamate concentration [89,90]. Excess intake at levels higher than normal human dietary consumption (as found in large pharmacological doses >12.0 g applied on an empty stomach without food) results in transiently elevated plasma levels [28,91,92]; which normalize within two hours after the offset of MSG consumption [80,93]. Ingesting MSG with food significantly blunts the plasma increase [93–97].

1.3. Glutamate and the human infant

The human amniotic fluid and breast milk contain significant levels of free natural glutamate and the infants have been shown to possess the capacity to effectively metabolize glutamate as the adult [24,28,80,87]. The placenta forms a natural barrier to the penetration of glutamate from maternal into fetal circulation. The role of placenta is actually more than that of a simple barrier, because it uses glutamate extensively as an energy substrate and extracts it from both maternal and fetal circulation to generate energy [98]. The infant liver also synthesizes glutamate from glutamine and exports it into the fetal circulation. This efficient metabolism explains why Stegink et al. [99] observed no increases in fetal plasma glutamate concentration when maternal plasma glutamate concentration was significantly increased by intravenous infusion. It has also been demonstrated that free glutamate concentration in breast milk does not rise when fasting mothers ingest pharmacological doses of MSG (150 mg/kg) that produce seven-fold increase in plasma glutamate concentration [100]. Hence, ingestion of high doses of MSG in maternal diet (which is unlikely in typical dietary consumption) will not expose an infant to free glutamate via breast milk. Indeed, JECFA [28] confirmed that dietary levels of added MSG and natural levels of glutamic acids in foods do not raise toxicological concerns in infants.

1.4. Glutamate and the human brain

Glutamate is the most abundant free amino acid in the brain due to its substantial role in learning and memory formation. The brain content of glutamate (12.0 μ mol/g) is much greater than the plasma concentration [101,102] preventing passive penetration of glutamate into the brain. Moreover, glutamate level in the extracellular fluid is not allowed to increase because of the existence of two transport systems (facilitative and active) in the brain. The facilitative transport system uses a saturable and stereo-selective mechanism to restrict glutamate to the luminal membrane. It actually allows glutamate to enter endothelial cells but does not support its passage from endothelial cells into the brain [103]. The active transport system of the blood brain barrier (BBB) expels glutamate from the extracellular fluid [102]. The gradient between the brain cells and the extracellular space is maintained by a sodium (Na^+) dependent glutamate transporter (excitatory amino acid transporters EAAT) that regulates the glutamate concentration in the extracellular space. The activity of the transporters allows glutamate to be recycled [85]. In situations of oxygen insufficiency/brain injury, the EAAT halts the neuro-transmission by sending excess glutamate to the outside cells [102]. The EAAT move glutamate from ECF into the endothelial cells where glutaminase hydrolyze and transforms glutamine to glutamate and ammonia (NH_4^+). The latter requires no carrier and is diffused as NH_3^+ with the remaining glutamine and glutamate into the blood [104].

The BBB participates in the regulation of brain nitrogen metabolism by preventing the accumulation of glutamate, glutamine and NH_4^+ [102]. Even in situations where plasma glutamate is increased ten-fold over an hour with a single dose of MSG (150 mg/kg BW) in a non-caloric liquid after an overnight fast (not typical in human dietary consumption), glutamate did not produce any effect on the human brain since no changes were observed in plasma concentration of pituitary hormones that are regulated by the hypothalamic glutamate synapses [105,106].

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2. MSG safety studies

The safety of any food substance depends on the purity level, route of administration, the quantity used, as well as other food components consumed with it. In scientific experiments, animal species used are typically mice/rats, dogs or monkeys. Studies on MSG revealed that mice were the most sensitive species to neonatal effects of MSG, especially when compared to primates [70].

Due to its extensive metabolism in the gastrointestinal tract, which was described Section 1.2 above, the routes of administration influences the character and extend of MSG effects. Consequently, subcutaneous or intravenous injection studies, as well as forced intubation, should not be extrapolated to normal oral use of MSG, because those routes overcome the metabolic pathway that exists for normally ingested glutamate. Studies that administered MSG through subcutaneous injection to animals produced abnormalities like retina and hypothalamus lesions [107], changes in hepatic fat metabolism [108–111] and brain injury [67,69,85,112,113]. Those studies provided for interesting pharmacological models but were irrelevant to food use and were unsuitable to human nutrition. Indeed, studies with MSG normally ingested [8,14,114–116] did not observe any hypothalamic damage, retinal lesions or hepatic fat pathologies. In line with the absence of those pathologies, a recent comprehensive review Brosnan et al. [78] concluded that MSG use was not associated with any form of human body weight change or obesity.

Not only the route, but also the quantity of any food substance determines its effect on consumption. The amount of glutamate needed to achieve brain lesion in mouse by gavage (approximately 0.5 g/kg BW) is not voluntarily achievable in humans, because it causes nausea [28].

As already mentioned (Section 1.2), the food components consumed together with MSG have either synergistic or antagonistic effects depending on their metabolic pathway. Among others, Fernstrom and Garattini [97] described that foods rich in carbohydrates reduce peak plasma concentration of glutamate. Barylko-Pikielna and Kostyra [117] illustrated that amino acids matrix in foods influences the flavor-enhancing effects of MSG and therefore MSG cannot be used indiscriminately to enhance taste of every food. Foods containing *trans*-fatty acids and fructose have long been implicated in abnormalities of fat metabolism in the liver and adipose tissue and some speculated that MSG may enhance those negative effects in mice [118–121]. However, other researchers were not able to replicate these results [116], which is understandable since MSG and *trans*-fats have completely different metabolic pathways

and thus a metabolic synergy is difficult to explain physiologically.

In addition, some reports [112,122,123] have associated MSG intake with adverse events like headaches, numbness, flushing, tingling, palpitations and drowsiness (Chinese restaurant syndrome) in some sensitive individuals consuming ≥ 3 g without food. Research experts [34,72,124,125] have not been able to consistently trigger reactions in double-blind studies with such individuals using MSG or placebo in food.

2.1. MSG safety and recent scientific articles

The growing use of prepackaged foods in the global community, as well as wide use of simple flavor enhancers in rural areas of the world, triggered a new interest of the academic community in MSG safety. In the recent decades, the following studies were conducted globally:

- Onaolapo et al. [50] used adult mice intubated with MSG at 10 mg/kg and found an increased plasma concentration of hormones, neuronal injury and oxidative stress.
- Ememali et al. [126] suggested alterations in brain serum enzyme activity, disruption of kidney functions and other hematological parameters following oral administration of 0.5–1.5 g/kg MSG to male rats for 4 wks using only distilled water as a control.
- Boonnate et al. [127] provided MSG 2 mg/g body weight to adult rats for 1–9 months and speculated negative effect on the pancreas.
- Ochiogu et al. [128] subcutaneously and orally administered MSG at 0.25–1.00 g/kg to albino rats for 6 wks. No control substance was provided. The author reported reductions in gonadotrophin-releasing hormone, luteinizing hormone, testosterone, total cholesterol and cauda epididymal sperm reserves.
- Sharm et al. [129] fed adult rats with MSG 2 mg/g body weight in drinking water and associated MSG intake with kidney injury
- Okon et al. [130] orally administered MSG at 4–8 mg/kg body weight to matured albino rats for 4 wks using normal saline as control and concluded significant increase in gross weight of heart.
- Shi et al. [47] concluded that MSG intake may have independent blood pressure elevation effects on women and those on hypertension medications
- Eweka et al. [131] forcedly administered 40–80 mg/kg MSG orally to 24 adult male Wistar rats and concluded that MSG was harmful to the liver.
- Ogbuonu et al. [132] forcedly administered MSG to adult rats at 15 mg/kg and suggested alteration in lipid status and damage to metabolic organs.
- Igwebuike et al. [133] and [134] used a gavage needle to administer MSG solution for 6 wks to male albino rats and reported a disruption of hypothalamic-pituitary-testes regulatory axis, as well as a lowered serum testosterone.
- Baad-Hansen et al. [135] associated oral MSG (75–150 mg/kg body weight) intake without food with headache, pericranial muscle sensitivity and elevated systolic blood pressure in humans
- Egbuonu et al. [136] intubated adult male rats with MSG for 4 wks concluded hepatotoxicity with no significant pathologies of the bone.
- Farombi and Onyema [137] intraperitoneally administered MSG to adult rats at 4.0 mg/g and concluded oxidative damage in the liver, kidney and brain.
- Mozes et al. [138] implicated MSG and overfeeding with obesity and changes in alkaline phosphatase activity in young and adult rats.
- Singh et al. [139] associated consumption of high dose (≥ 4 mg/g body weight) MSG with oxidative stress in the hepatic tissue of adult mice

It is not the purpose of this review to comment on pharmacological usefulness of the above rodent models. However, virtually all authors listed above made far-reaching interpretations of their results to normal human food use of MSG without discussing species difference, dosing, absence of positive controls for sodium and animal model relevance.

Noticeably, mice are substantially more vulnerable to MSG neonatal effects than dogs or primates [70] due to the different evolution of the blood-brain barrier. Moreover, forced injection and intubation of MSG solution to an empty stomach is not typical to human dietary consumption and those routes prevent normal gastrointestinal metabolism of glutamate. This is best illustrated by the increase in plasma glutamate reported by Onaolapo et al. [50] which would have not been achievable was MSG ingested normally by rats. The study has been extensively criticized for other methodological problems related to brain dissection and plasma sample collection [140].

The typical dietary intake level of MSG worldwide is in the range of 0.6–1.5 g/person/day [40,43,63,76,77], which is a tiny fraction of the total dietary daily glutamate intake from natural sources [74,76,141]; Thus, it is hard to attribute causal role in major central and endocrine pathologies to only a single source of dietary glutamate. Indeed, the above-listed studies have not explicably identified a causal mechanism by which MSG could selectively produce the indicated results. The authors did not account for the fact that glutamate was metabolically compartmentalized and did not cross biologic membranes [80–88,98,100].

Furthermore, there is scientific evidence demonstrating the activities of the transport systems that maintain the brain concentration of glutamate; the dose of glutamate required at the target site to lead to continuous excitation of the glutaminergic neurons, depleting ATP and leading to cell death [28,70], and the inability of dietary use of MSG to disrupt human neuro-endocrine axis [33,106].

No physiological mechanism by which ingested MSG could selectively influence hepatic and adipose tissue fat synthesis has been described considering its extensive metabolism in the gastrointestinal tract. MSG doses used in some of the above studies are negligible when compared to MSG from dietary sources [142] and the normal concentration of hepatic glutamate (≤ 2.5 mM) cannot activate the key enzymes responsible for fatty acid biosynthesis in the liver (acetyl-CoA carboxylase) [143] since the enzyme requires a K_m of 25 mM for the activation [144,145]. Although marked increase in plasma glutamate concentration can cause a modest increase in plasma insulin concentration and induce fatty acid synthesis [103] typical dietary MSG intake cannot cause significant increase in insulin plasma concentrations [146].

Finally, several toxicological and epidemiological studies have reported an absence of effects of MSG on body weight in animals or humans [147–158]. None of these studies were referred in the research conducted in Nigeria.

3. Conclusion

The recent articles published by several scientists, which speculatively linked animal pharmacology to human food use of MSG, are inconsistent with already established evidence-based scientific findings. MSG consumed with food does not present any of the incriminating disease conditions, such as brain effects or hormonal imbalances. Vice versa, ingestion of MSG as a dietary constituent,

drives food intake of protein foods and supports some physiological functions, such as intestinal motility. Any future research on MSG should consider glutamate metabolism and its compartmentalization in various organs.

Conflict of interests

None, the author is a registered dietitian and an international nutritionist specializing in food policy.

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